

Sandesh Bhandari

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Research Interests

Statistical and machine learning approaches to urban spatial data — how cities can be modeled, predicted, and understood through geographic and earth-systems data at fine spatial scale. My focus is on problems where the spatial relationships between observations are fundamental to how the data should be modeled, explored, and interpreted.

Education

University of Toronto, Faculty of Arts and Science

Incoming, September 2026

Master of Science, Department of Geography and Planning

University of Toronto, Faculty of Arts and Science

2026

HBSc. in Geography, Earth and Data Sciences

with High Distinction

- GPA: 3.92/4.0 · Dean's List Scholar
- Nicholas Wemyss Undergraduate Explorers Fund Award
- John Horner Undergraduate Scholarship in Geography
- University of Toronto Research Excellence Award (UTEA)
- Undergraduate Computer Applications Award

Experience

City of Toronto, Planning Research and Analytics

May 2026 – Present

Research Assistant

- Conducting field data collection and verifying the accuracy of site records.
- Structuring and presenting collected data to support project reporting.
- Maintaining statistical databases to ensure data integrity.
- Gathering employment and land-use data from Toronto businesses to inform municipal planning and future city services.

University of Toronto, Department of Geography & Planning

January 2026 – Present

Research Assistant

- Supporting the primary research mandate through exploratory spatial analysis and rapid data visualization to iteratively test hypotheses and validate assumptions.
- Conducting spatial analysis of residential zoning typologies (R, RD, RS, RT, RM) along major arterial corridors to evaluate densification potential under updated Toronto zoning policies.
- Developing GIS workflows to classify parcel frontage (Front, Flank, Reverse) and model transit catchment zones relative to Major Transit Station Areas (MTSAs).

Web Developer

2019 – Present

Independent Contractor

- Architected and deployed 15+ full-stack web applications using MERN/MEAN stacks (MongoDB, Express, React/Angular, Node.js) for small-to-medium enterprises.
- Optimized front-end performance by migrating legacy CSS to SCSS and implementing Alpine.js for lightweight state management, reducing bundle sizes significantly and improving Lighthouse scores.
- Integrated secure payment gateways (Stripe, PayPal) and third-party REST APIs.
- Managed end-to-end project lifecycles, negotiating contract scopes, setting technical roadmaps, and delivering 100% of client projects on or ahead of schedule.

Skills

GIS & Spatial Analysis

ArcGIS, QGIS, GeoPandas, Rasterio, Shapely, GDAL/OGR, OSMnx, PySAL, NetworkX, Earth Engine

Programming & Data

Python, JavaScript, TypeScript, C, Rust, SQL, Bash, NumPy, SciPy, Pandas, Polars, Node, Tauri

AI & Machine Learning

PyTorch, PyTorch Lightning, scikit-learn, LightGBM, SHAP, Hugging Face Transformers, semantic segmentation (SegFormer), OpenCV

Design & Visualization

D3.js, THREE.js, React Three Fiber, GLSL/WebGL, Canvas 2D, Leaflet, Folium, Matplotlib, Seaborn, Figma, Adobe